

MoS₂, DFT plane waves:

NBND	350	320	270	220	170	120	60
m_{V-1}	-3.95	-3.94	-3.92	-3.86	-3.72	-3.51	-3.20
m_V	-4.16	-4.16	-4.14	-4.09	-3.94	-3.74	-3.43
m_C	-2.05	-2.07	-2.02	-2.06	-2.08	-2.04	-1.99
m_{C+1}	-1.82	-1.84	-1.79	-1.84	-1.85	-1.81	-1.75

MoS₂, Siesta converted to plane wave [numerical e-16j imaginary part]:

- Overestimated berry curvature around 10%

NBND	90	80	70	60
m_{V-1}	-3.68	-3.58	-3.50	-3.51 (11%)
m_V	-4.05	-3.97	-3.88	-3.89
m_C	-2.45 (19%)	-2.42	-2.32	-2.33
m_{C+1}	-1.75	-1.73	-1.60	-1.63

MoS₂, Siesta extracting Hamiltonian [numerical e-09j]:

- Used generalized eigenvalue method, i.e. overlap integrals $\langle u_{nk} | u_{nk+dk} \rangle = u_{nk}^\dagger S_k u_{nk+dk}$

NBND	90	80 (removing part of H, S)
m_{V-1}	-2.75	-2.58
m_V	-2.96	-2.76
m_C	-1.91	-1.65
m_{C+1}	-1.72	-1.46

- For siesta -> plane wave, m_C is quite off. However, the other values seem like they will converge to reasonably accurate g-factors.
- For siesta -> Hamiltonian, the conduction g-factors seem reasonable, slightly underestimated. The valence ones are much lower than expected, however.
- Berry factors had errors (~.1% for valence) (~10% for conduction) for the Hamiltonian method

WSe₂, DFT plane waves:

NBND	350	320	270	220	170	120	60
m_{V-1}	-4.04	-4.04	-4.01	-4.00	-3.86	-3.66	-3.23
m_V	-4.87	-4.87	-4.85	-4.84	-4.71	-4.55	-4.12
m_C	-1.86	-1.87	-1.84	-1.86	-1.85	-1.91	-1.84
m_{C+1}	-2.86	-2.87	-2.84	-2.85	-2.83	-2.91	-2.78

WSe₂, Siesta converted to plane waves [numerical e-16j]:

NBND	110	90	75	60
m_{V-1}	-3.92	-3.63	-3.34	-3.21
m_V	-4.95	-4.67	-4.36	-4.20
m_C	-1.91	-1.85	-1.73	-1.77
m_{C+1}	-3.11 (9%)	-3.03	-2.95	-2.75

WSe₂, Siesta extracting Hamiltonian [numerical e-09j]:

NBND	110	100 (removing part of H, S)
m_{V-1}	-2.98	-2.92
m_V (off, but m_{V-1} , m_V correct ratio)	-3.58	-3.51
m_C	-1.73	-1.35
m_{C+1}	-2.61	-2.18

- For siesta -> plane wave, values are generally reasonable. Not sure what the accuracy would be when using more bands (different PAO Basis)
- For siesta -> Hamiltonian, the conduction g-factors seem reasonable, slightly underestimated. The valence ones are much lower than expected, same case as MoS₂.
- Berry factors had errors (~2% for valence) (~10% for conduction) for the Hamiltonian method

